

Clearly, in some cases, the solution is trivial. The sorting robots of Flipkart or the US Postal Service don't really need to go anywhere - they can be plugged into the wall, and that will be that. But what about quadcopters, UAVs (unmanned aerial vehicles), exploratory robots that search for oil or underwater ruins? What about satellites, surveillance bots, or even simple high-powered robots that need to work in remote areas?

Batteries have long been the go-to option whenever a power socket or fuel generator was unavailable. Depending on power requirements, weight, and recharge requirements, types of batteries like alkaline, lithium ion and nickel-metal hydride batteries have gone into mass production with a host of diverse applications. Novel methods such as solar cells, biofuels and geothermal power from the ground, or vibrational energy from a superstructure have been harnessed to recharge these batteries on a regular basis. However, in many electrical circuits, a battery is simply not up to the mark - its longevity, power, and consistency are constantly thrown in doubt.

Pneumatic and hydraulic power are the solutions to this quandary, albeit with the limitation of infrastructure. With a fully functional pipeline supplying high-pressure air or water, the possibilities are nearly endless. Nearly any kind of control system can be implemented using pneumatic circuits, the components of which are valves and switches, quite akin to the gates and switches seen in digital circuits.

What controls it - microprocessors

These aren't the clunky valve-transistor relics from the 80s. Nor are they the massive motherboards inside your CPUs from the turn of the millennium. These are slick, robust calculation monsters, designed to fit in areas as small as the palm, with diverse yet powerful I/O, memory management and programming abstractions. Control boards like the Arduino and its various knockoffs, Raspberry Pi, as well as custom boards built by the likes of Intel



Source: Popsai

An automated toaster using a simple microcontroller schema.